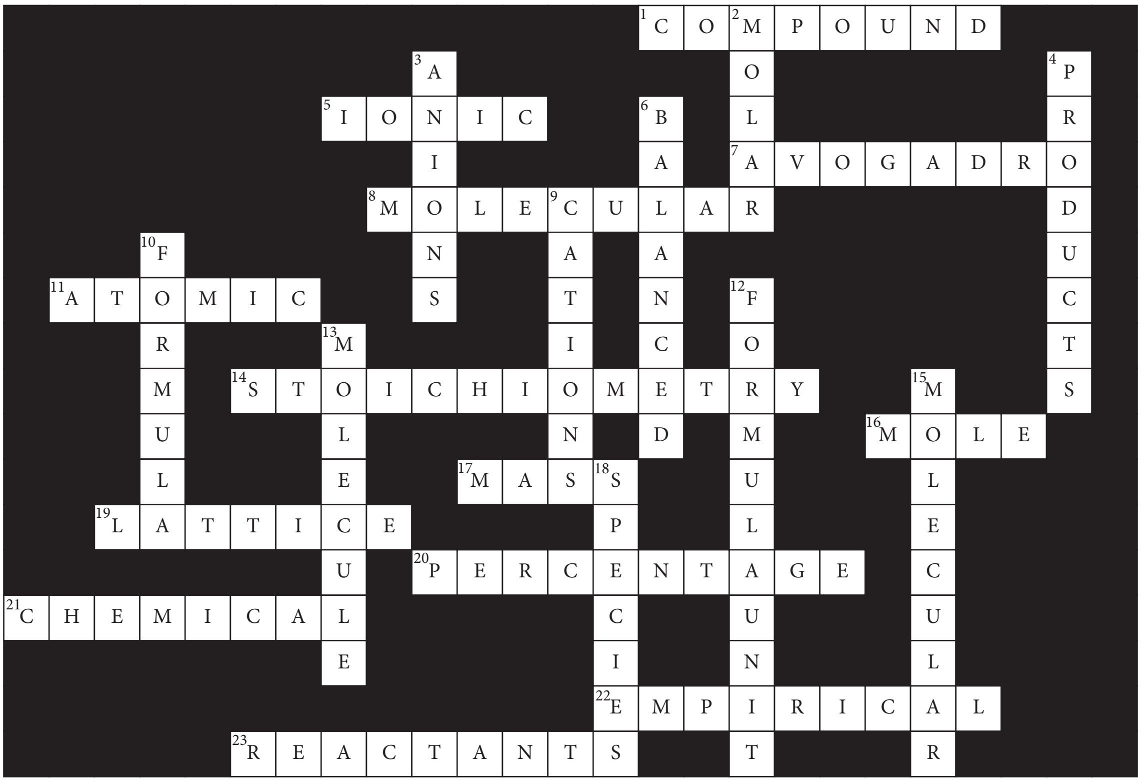
ACTIVITY SHEET ANSWERS

Chapter 5 Suggested answers

Prior knowledge

1 nucleus
2 mass
3 electrons
4 mass number
5 atomic number
6 element
7 symbol
8 neutrons
9 isotopes
10 numbers
11 neutral
12 stable
13 carbon-12
14 relative
15 standard
16 atomic mass
17 periodic
18 abundance
19 hydrogen
20 oxygen

Activity sheet 5.1



Activity sheet 5.2

Part A

1 $A_r(\text{Fe}) = 55.85$

2 $A_r(\text{H}^+) = A_r(\text{H}) = 1.01$

$A_r(\text{Cl}^-) = A_r(\text{Cl}) = 35.45$

3 $M_r(\text{H}_2) = 2 \times A_r(\text{H}) = 2 \times 1.01 = 2.02$

4 $\text{Fe(s)} + 2\text{H}^+ + 2\text{Cl}^- \rightarrow \text{Fe}^{2+} + 2\text{Cl}^- + \text{H}_2\text{(g)}$

5 1 mol + 2 mol + 2 mol → 1 mol + 2 mol + 1 mol

6 2 mol of H^+ reacts with 1 mol of Fe, so 4 mol of H^+ reacts with 2 mol of Fe

7 $n(\text{Fe}) = 2$ mol, $M(\text{Fe}) = 55.85$ g mol^{-1}

$m(\text{Fe}) = n(\text{Fe}) \times M(\text{Fe}) = 2 \times 55.85 = 111.7$ g

8 2 mol of H^+ produces 1 mol of H_2, so 4 mol of H^+ produces 2 mol of H_2

9 $n(\text{H}_2) = 2$ mol, $M(\text{H}_2) = 2.02$ g mol^{-1}

$m(\text{H}_2) = n(\text{H}_2) \times M(\text{H}_2) = 2 \times 2.02 = 4.04$ g

Part B

1 $A_r(\text{Ca}) = 40.08$

$A_r(\text{H}^+) = 1.01$

2 $M_r\left(\text{So}_4^{2-}\right) = A_r(\text{S}) + 4 \times A_r(\text{O}) = 32.06 + 4 \times 16.00 = 32.06 + 64.00 = 96.06$

$M_r(\text{CaSO}_4) = A_r(\text{Ca}) + A_r(\text{S}) + 4 \times A_r(\text{O}) = 40.08 + 32.06 + 4 \times 16.00 = 136.14$

3 $M_r(\text{H}_2) = 2 \times A_r(\text{H}) = 2 \times 1.01 = 2.02$

4 $\text{Ca(s)} + 2\text{H}^+ + \text{So}_4^{2-} \rightarrow \text{CaSO}_4\text{(s)} + \text{H}_2\text{(g)}$

5 1 mo + 2 mol + 1 mol → 1 mol + 1 mol

6 2 mol of H^+ reacts with 1 mol of Ca, so 0.5 mol of H^+ reacts with 0.25 mol of Ca

7 $n(\text{Ca}) = 0.25$ mol, $M(\text{Ca}) = 40.08$ g mol^{-1}

$m(\text{Ca}) = n(\text{Ca}) \times M(\text{Ca}) = 0.25 \times 40.08 = 10.02$ g

8 2 mol of H^+ produces 1 mol of H_2, so 0.5 mol of H^+ produces 0.25 mol of H_2

9 $n(\text{H}_2) = 0.25$ mol, $M(\text{H}_2) = 2.02$ g mol^{-1}

$m(\text{H}_2) = n(\text{H}_2) \times M(\text{H}_2) = 0.25 \times 2.02 = 0.51$ g

Activity sheet 5.3

1 a $M_r(\text{NH}_3) = A_r(\text{N}) + 3 \times A_r(\text{H})$

$= 14.01 + 3 \times 1.01$

$= 17.04$

b $M_r(\text{H}_2\text{SO}_4) = 2 \times A_r(\text{H}) + A_r(\text{S}) + 4 \times A_r(\text{O})$

$= 2 \times 1.01 + 32.06 + 4 \times 16.00$

$= 98.08$

c $M_r(H_2CO_3) = 2 \times A_r(H) + A_r(C) + 3 \times A_r(O)$

$= 2 \times 1.01 + 12.01 + 3 \times 16.00$

$= 62.03$

2 **a** $M_r(Na_2SO_4) = 2 \times A_r(Na) + A_r(S) + 4 \times A_r(O)$

$= 2 \times 22.99 + 32.06 + 4 \times 16.00$

$= 142.04$

b $M_r(Mg(NO_3)_2) = A_r(Mg) + 2 \times A_r(N) + 6 \times A_r(O)$

$= 24.31 + 2 \times 14.01 + 6 \times 16.00$

$= 148.33$

c $M_r(Mg(OH)_2) = A_r(Mg) + 2 \times A_r(O) + 2 \times A_r(H)$

$= 24.31 + 2 \times 16.00 + 2 \times 1.01$

$= 58.33$

3 **a** mass of Mg = 24.31

mass of Cl = 2 × 35.45 = 70.90

formula mass of $MgCl_2$ = mass of Mg + mass of Cl

= 24.31 + 70.90 = 95.21

% Mg = 24.31 × 100/95.21 = 25.5%

% Cl = 70.90 × 100/95.21 = 74.5%

b mass of Ca = 40.08

mass of C = 12.01

mass of O = 3 × 16.00 = 48.00

formula mass of $CaCO_3$ = mass of Ca + mass of C + mass of O

= 40.08 + 12.01 + 48.00 = 100.09

% Ca = 40.08 × 100/100.09 = 40.0%

% C = 12.01 × 100/100.09 = 12.0%

% O = 48.00 × 100/100.09 = 48.0%

c mass of Al = 2 × 26.98 = 53.96

mass of S = 3 × 32.06 = 96.18

mass of O = 12 × 16.00 = 192.00

formula mass of $Al_2(SO_4)_3$ = mass of Al + mass of S + mass of O

= 53.96 + 96.18 + 192.00 = 342.14

% Al = 53.96 × 100/342.14 = 15.8%

% S = 96.18 × 100/342.14 = 28.1%

% O = 192.00 × 100/342.14 = 56.1%

Activity sheet 5.4

1 **a** $n(\text{He})$ = number of particles/N_A

$= 3.01 \times 10^{23}/6.02 \times 10^{23}$

$= 0.05 \times 10^{23-23}$

$= 0.05 \times 10^{0} = 0.05$ mol

b $n(CO_2)$ = number of particles/N_A

$= 12.04 \times 10^{24}/6.02 \times 10^{23}$

$= 2.00 \times 10^{24-23}$

$= 2.00 \times 10^{1} = 20.0$ mol

c $n(Cu(NO_3)_2)$ = number of particles/N_A

$= 18.06 \times 10^{22}/6.02 \times 10^{23}$

$= 3.00 \times 10^{22-23}$

$= 3.00 \times 10^{-1} = 0.30$ mol

2 **a** number of particles = $n(\text{C}) \times N_A$

$= 2.50 \times 6.02 \times 10^{23}$

$= 15.1 \times 10^{23}$

$= 1.51 \times 10^{24}$

b number of particles = $n(SO_2) \times N_A$

$= 0.01 \times 6.02 \times 10^{23}$

$= 0.060 \times 10^{23}$

$= 6.0 \times 10^{21}$

c number of particles = $n(\text{So}_4^{2-}) \times N_A$

$= 12.5 \times 6.02 \times 10^{23}$

$= 75.3 \times 10^{23}$

$= 7.53 \times 10^{24}$

3 **a** 1 mol of H_2SO_4 contains 2 mol H atoms, 1 mol S atoms, and 4 mol O atoms, so 4.0 mol of H_2SO_4 contains 8.0 mol H atoms, 4.0 mol S atoms, and 16.0 mol O atoms

b total number of moles of atoms = 4.0 + 8.0 +16.0 = 28.0 mol

c number of atoms = $28.0 \times 6.02 \times 10^{23} = 169 \times 10^{23} = 1.69 \times 10^{25}$

4 **a** 1 mol of $CaSO_4$ contains 1 mol Ca atoms, 1 mol S atoms, and 4 mol O atoms, so 0.5 mol of $CaSO_4$ contains 0.5 mol Ca atoms, 0.5 mol S atoms, and 2.0 mol O atoms

b total number of moles of atoms = 0.5 + 0.5 + 2.0 = 3.0 mol

c number of atoms = $3.0 \times 6.02 \times 10^{23} = 18 \times 10^{23} = 1.8 \times 10^{24}$

5 **a** 1 mol of $Al_2(SO_4)_3$ contains 2 mol Al atoms, 3 mol S atoms, and 12 mol O atoms, so 17.5 mol of $Al_2(SO_4)_3$ contains 35.0 mol Al atoms, 52.5 mol S atoms, and 210 mol O atoms

b total number of moles of atoms = 35.0 + 52.5 + 210 = 298 mol

c number of atoms = $298 \times 6.02 \times 10^{23} = 1790 \times 10^{23} = 1.79 \times 10^{26}$

Chapter 5 Revision

1 a → L, b → S, c → D, d → N, e→ O, f → Q, g → Y, h → F, i → V, j → E, k → B, l → T, m → R, n → J, o → A, p → C, q → G, r → H, s → P, t → M, u → W, v → I, w → U, x → K, y → X

2 **a** *Relative atomic mass* refers to the mass of an atom on a scale in which the mass of an atom of the carbon-12 isotope is exactly 12. *Relative molecular mass* refers to the mass of a molecule on a scale in which the mass of an atom of the carbon-12 isotope is exactly 12. *Relative formula mass* refers to the mass of one formula unit of an ionic compound on a scale in which the mass of an atom of the carbon-12 isotope is exactly 12.

b An *empirical formula* is the simplest whole number ratio of the different atoms found in a compound. A *molecular formula* specifies the actual number of each type of atom present in a molecule.

c A *mole* of a substance is the amount of a substance that contains 6.02×10^{23} particles of that substance. The *molar mass* of a substance is the mass in grams of one mole of a substance.

3 **a** False (to a *carbon-12* atom)

b True

c False (*Some* pure substances)

d True

e True

f False (6.02×10^{23} particles)

g True

h False (measured in grams)

i True

j True

4

Quantity	Symbol	Units	Unit symbols
Relative atomic mass	A_r	atomic mass units	amu
Relative molecular mass	M_r	*atomic mass units*	*amu*
Relative formula mass	M_r	*atomic mass units*	*amu*
Mass	m	*gram*	*g*
Number of moles	n	*mole*	*mol*
Molar mass	M	*grams per mole*	$g\ mol^{-1}$

5 Step 1 Identify the number of each type of atom.

Step 2 Multiply the number of each type of atom by its relative atomic mass (from periodic table).

Step 3 Add the masses of each element to obtain the relative molecular mass of the molecular compound.

6 Step 1 Identify the number of each type of atom.

Step 2 Multiply the number of each type of atom by its relative atomic mass (from periodic table).

Step 3 Add the masses of each element to obtain the relative formula mass of the ionic compound.

7 Step 1 Calculate the relative mass of each element in the compound.

Step 2 Calculate the relative formula or molecular mass of the compound.

Step 3 Calculate the percentage of each element by mass in the compound.

8 Divide the number of particles specified by the Avogadro constant (N_A).

9 Multiply the number of moles specified by the Avogadro constant (N_A).

10 Add the unit gram (g) to its relative atomic, molecular or formula mass.

11 Step 1 Write the relationship, $n = m/M$.

Step 2 Calculate the molar mass (M) of the element or compound.

Step 3 Substitute the molar mass (M) and the given mass (m) into the formula and complete the calculation to give the number of moles (n).

12 Step 1 Rearrange the formula $n = m/M$ to give $m = n \times M$.

Step 2 Calculate the molar mass (M) of the element or compound.

Step 3 Substitute the number of moles (n) and the molar mass (M) into the formula and complete the calculation to give the actual mass (m).

13 Step 1 Write a balanced equation and determine the molar ratio for the reaction from the coefficients of the species involved.

Step 2 Use the molar ratio to calculate the number of moles of the required reactant or product given the specified amount of another reactant or product.